CHAPTER ONE

DIFFERENTIAL CALCULUS OF FUNCTIONS OF SEVERAL VARIABLES

1: DEFINITIONS AND EXAMPLES OF FUNCTION OF SEVERAL VARIABLES

Definition(Function of two variables)

Let D be the non-empty subset of xy-plane. A function of two variables x and y is a rule that assign a unique real number $f(x,y)$ to each point (x,y) in some subset D of $\mathbf{R^2}$.

Definition(Function of three variables)

Let D be the non-empty subset of three dimensional spaces. A function of three variables x, y and z is a rule that assign a unique real number $f(x,y,z)$ to each point (x,y,z) in some subset D of $\mathbf{R^3}$.

Note that:

1. The domain of function of several variables is the set of all possible value of the variables for which the function is defined.
2. The range of f is set of all values assumed by f
   i.e. $R_f = \{z / z=f(x, y) \text{ form some } (x, y) \text{ in } D\}$.

Example1: Find the domain of the following

1. $f(x,y)=\sqrt{1-x^2-y^2}$, domain of f is $\{(x,y): x^2+y^2 \le 1\}$.
2. $f(x,y)=\dfrac{xy}{x^2+y^2}$, domain of f is $\{(x,y):(x,y)\neq(0,0)\}$
3. $f(x,y)=\sqrt{y+1}+\ln(x^2-y)$ domain of f is $\{(x,y)\in R^2 / -1\le y<x^2\}$.
4. $f(x,y)=\dfrac{1}{\sqrt{xy}}$, domain of f is $\{(x,y): xy>0\}$.

Example2: Find the domain of $f(x,y,z)=\sqrt{1-x^2-y^2-z^2}$.

Then domain of f is given by $\{(x,y,z): x^2+y^2+z^2 \le 1\}$ Which is a ball (a solid sphere) of radius 1

2. The graph of function of two variables.

The graph of $f(x,y)$ is the set of all points in $\mathbf{R^3}$ given

Graph of $f = \{(x,y,z)/\ z=f(x,y),(x,y)\in D_f\}$.

The graph of functions of two variables is a **Surface** can be sketched in $\mathbf{R^3}$.

Example 3: Sketch the graph of $f(x,y)=\sqrt{1-x^2-y^2}$.

Solution: $z=\sqrt{1-x^2-y^2} \Leftrightarrow z^2+x^2+y^2=1$, i.e. the graph of f is the set of points in R^3 which are located 1 unit from $(0,0,0)$ and above the xy plane. So it is a hemisphere.

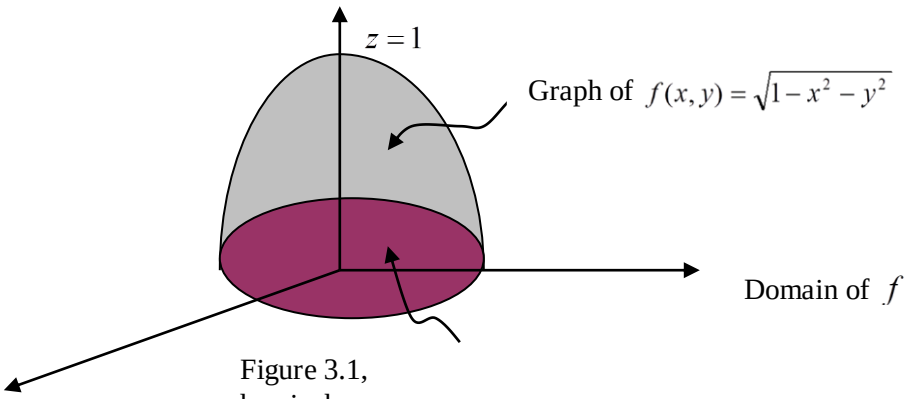
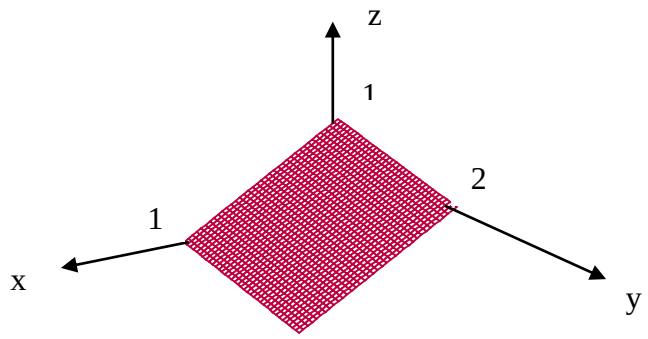


Figure 3.1, hemisphere

Example 4: Sketch the graph of $f(x,y)=1-x-\frac{y}{2}$.

Solution: Let $z=f(x,y)$.Then $z=1-x-\frac{y}{2}\Leftrightarrow x+\frac{y}{2}+z=1$.We can see that the intersection of the graph of f with the xy plane is a line $x+\frac{1}{2}y=1$. Moreover, the intersection of the graph of f and xz plane is a line $x+z=1$. Finally the intersection of the graph of f with the yz plane is $\frac{1}{2}y+2=1$, i.e. we have the following graph.

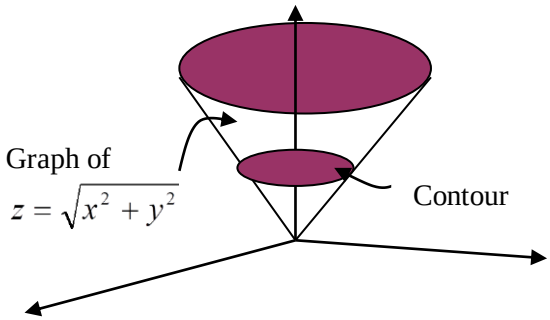


Example 5: Sketch the graph of $z=\sqrt{x^2+y^2}$

Solution: Clearly, $z\ge 0$.Then, the graph is expected to be above the xy plane. Now, we find the intersection of the graph of $f(x,y)=\sqrt{x^2+y^2}$ with the planes $z=0, y=0$ and $x=0$. Then,

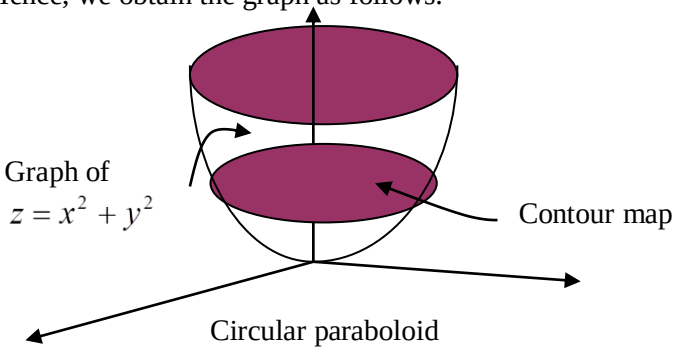
(i) the intersection of the graph with $z=0$ (xy plane) is the point (0, 0).

(ii) the intersection of the graph with $x=0$ is two intersecting lines $z=y$ and $z=-y$.The intersection of the graph with $y=0$ is $z=x$ and $z=-x$.Moreover, for any $k>0$, $z=k$ intersects the graph $z=\sqrt{x^2+y^2}$,i.e. $x^2+y^2=k^2$ at a circle of radius $r=k$ with center (0, 0).Thus, contour lines are circles of radius r. On the other hand, the intersection of the graph of $f(x,y)=\sqrt{x^2+y^2}$ with the co-ordinate axes is intersecting lines passing through the point $(0,0)$.



Example 6: Sketch the graph of $z = x^2 + y^2$.

Solution: The intersection of the graph of $z = x^2 + y^2$ and the co-ordinate planes xz, yz and xy planes are the parabola $z = x^2, z = y^2$ and the point $(0,0)$ respectively. The contour maps for any $k \geq 0$ is either a point or a circle. Moreover, since the value of $z \geq 0$, the graph is above the xy plane. Hence, we obtain the graph as follows.



3. LEVEL CURVES

One method of visualizing function of two variables is looking at its graph. Another method, borrowed from map markers, is a Contour map on which points of a constant elevation are joined to form level curve (contour curve).

Suppose $z = f(x, y)$ is a function of two variables and let $z = c$, $c \in \mathrm{R}$ be a plane. Then, the intersection of the graph of $z = f(x, y)$ and the plane can be seen to be $f(x, y) = c$. One can see that this is obtained by cutting the graph of $z = f(x, y)$ by the plane $z = c$. Then, the resulting map (intersection) is called the contour map corresponding to $z = c$. The projection of this contour map on the xy plane is called level curve (contour curve) with constant k.

It is generally hard to draw a picture of the graph of a function of two variables. Sometimes, But, we can sketch level curves. Take a function f of two variables x and y, defined on a domain D. For a real number k, the level curve at k, denoted $C(k)$ is the set of all points in D such that $f(x, y) = k$, i.e. $c(k) = \{(x, y) \in D : f(x, y) = k\}$. The following figures illustrates these two terms.

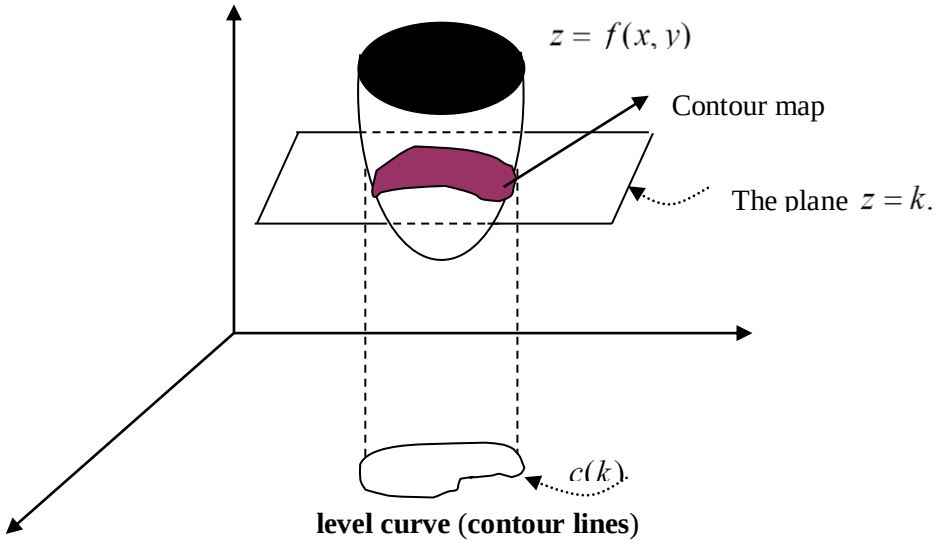
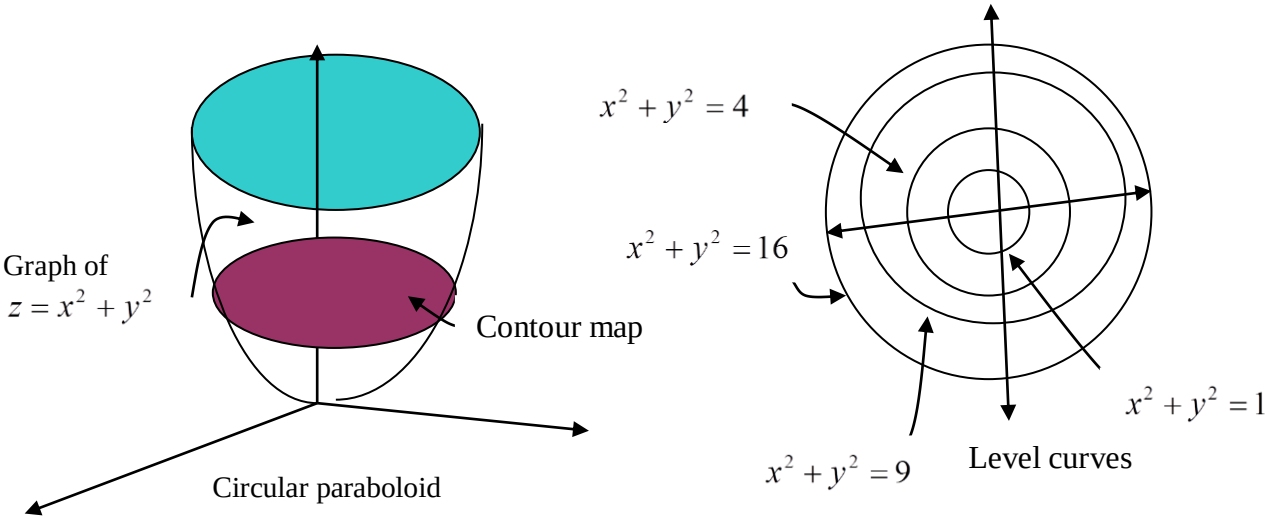


Figure 3.3

Example 7 Let $f(x, y) = x^2 + y^2$.

- If $k < 0$, then $c(k)$ is empty set.
- $c(0)$ contains a single point, namely the origin.
- If $k > 0$, then $c(k)$ is a circle centered at the origin and whose radius is equal to $\sqrt{k}$.



4: QUADRIC SURFACES

Definition Let $w = f(x, y, z)$ be a function of three variables. Then, the set of points (x, y, z) in $\mathbf{R}^3$ such that $f(x, y, z) = k$ for appropriately chosen real number k, is called the **level surface** with respect to the constant k.

Example 8: the level surface of

a. $f(x,y,z)=x^2+y^2+z^2 \quad \Rightarrow x^2+y^2+z^2=k\ ,\ k\geq 0$ *is a sphere.*

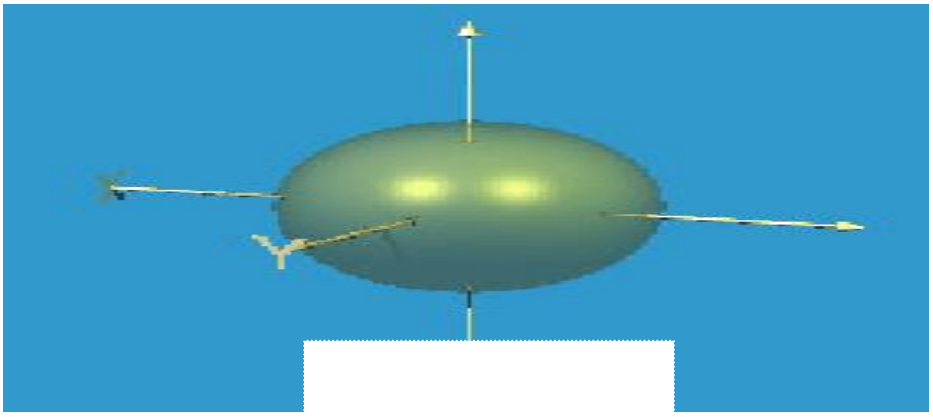
b. $f(x,y,z)=x^2+y^2 \quad \Rightarrow x^2+y^2=k\ ,\ k\geq 0$ *is a Cylinder.*

c. $f(x,y,z)=ax+by+cz \quad \Rightarrow ax+by+cz=k$ *it is a planes.*

Definition: surfaces defined by polynomial of the form are called **Quadric surface**.

A: ELLIPSOID

The surface whose equation is $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ where *a*, *b* and *c* are given positive is called an **ellipsoid**. Its intersection with any plane, parallel to coordinate planes, is an ellipse. If $a=b=c$, then we see that the ellipsoid is a sphere (see the figure below)



Example 9: a) $x^2+4y^2+z^2=4$

b) $9x^2+4y^2+z^2=1$

B: ELLIPTICAL CYLINDER

The surface with equation $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is called an elliptic cylinder. If in particular, $a=b$, we get $x^2+y^2=a^2$ which is called a **circular cylinder**.

Example 10: $y^2+9z^2=9$

C: ELLIPTICAL PARABOLOID

The surface whose equation is $z=ax^2+by^2$ where *a* and *b* are given positive numbers, is called a paraboloid.

Example 11: $x^2+4y^2-z=0.$

D: HYPERBOLIC PARABOLOID

The surface whose equation is $\frac{x^2}{a^2}-\frac{y^2}{b^2}=\frac{z}{c}$ where *a*, *b*, *c* are given positive numbers, is called a **hyperbolic paraboloid**.

Example 12 $z=2x^2-y^2$.

E: ELLIPTICAL DOUBLE CONE

The surface whose equation is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{z^2}{c^2}$ where *a*, *b*, *c* are given positive numbers, is called **double cone**.

Example 13 $16x^2=y^2+4z^2$.

5: LIMIT OF FUNCTIONS OF TWO /THREE/VARIABLES

Definition: Let f be a function defined throughout a set containing a disk centered at (x_o,y_o) except possibly at (x_o,y_o) itself, and let L be a real number. L is the limit of f at (x_o,y_o) if for every $\in>0$ *their exists a* $\delta>0$ such that

$$0<\sqrt{(x-x_o)^2+(y-y_o)^2}<\delta \Rightarrow |f(x,y)-L|<\in \Leftrightarrow \lim_{(x,y)\mapsto(x_o,y_o)} f(x,y)=L.$$

Example 14: Show that $\lim_{(x,y)\to(x_0,y_0)}(x+y)=x_0+y_0$ using $\varepsilon-\delta$ definition.

Proof:

Let $\varepsilon>0$ be given. For all $(x,y)\in R^2$, we can easily see that $|x-x_0|\le\sqrt{(x-x_0)^2+(y-y_0)^2}$ and $|y-y_0|\le\sqrt{(x-x_0)^2+(y-y_0)^2}$ .Moreover, we can see that

$$\begin{aligned}|(x+y)-(x_0+y_0)| &= |(x-x_0)+(y-y_0)|\\ &\le |x-x_0|+|y-y_0|\\ &\le 2\sqrt{(x-x_0)^2+(y-y_0)^2}\end{aligned}$$

Choose $\delta\le\frac{\varepsilon}{2}$. Then, it follows that

$$|(x+y)-(x_0+y_0)|=|(x-x_0)+(y-y_0)|\le|x-x_0|+|y-y_0|<2.\frac{\varepsilon}{2}=\varepsilon$$

whenever $0<\sqrt{(x-x_0)^2+(y-y_0)^2}<\delta,(x,y)\in R^2$.

Thus, we conclude that $\lim_{(x,y)\to(x_0,y_0)}(x+y)=x_0+y_0$.

Theorem.1: The limit of a function of two variables (if exists) is unique.

Theorem 2: Let $P=(x,y),P_0=(x_0,y_0)$ and suppose $\lim_{P\to P_0} f(P)=a$, and $\lim_{P\to P_0} g(P)=b$.

(a) $\lim_{lP\to P_0}(f\pm g)(P)=a\pm b$ (c) $\lim_{P\to P_0}(f.g)(P)=a.b$

(b) $\lim_{P\to P_0}(cf)(P)=ca$ (d) $\lim_{P\to P_0}(\frac{f}{g})(P)=\frac{a}{b},\ g(P)\neq 0, b\neq 0.$

Example: 15: Consider the function $f(x,y)=\dfrac{x\sin y+x^2+2}{\sqrt{x^2+y^2+1}}$. Then,

$$\lim_{(x,y)\to(0,0)} f(x,y)=\lim_{(x,y)\to(0,0)}\frac{x\sin y+x^2+2}{\sqrt{x^2+y^2+1}}$$

$$=\frac{\lim_{(x,y)\to(0,0)}\left(x\sin y+x^2+2\right)}{\lim_{(x,y)\to(0,0)}\sqrt{x^2+y^2+1}} \text{ (by quotient rule)}$$

$$=2.$$

Example 16: Show that $\lim_{(x,y)\to(0,0)}\dfrac{xy}{x^2+y^2}$ doesn't exist.

Solution: on the board.

Example 17: Compute the limit at (0, 0) of the function *f* such that

$$f(x,y)=\frac{xy-y^2}{\sqrt{x}-\sqrt{y}}.$$

Solution: on the board.

Example 18:

(i) Let $f(x,y)=x^2y+2xy^3-1$. Then

$$\lim_{(x,y)\to(2,1)}\left(x^2y+2xy^3-1\right)=\lim_{(x,y)\to(2,1)}(x^2y)+\lim_{(x,y)\to(2,1)}(2xy^3-1)=4+3=7$$

(ii) Let $f(x,y)=\sqrt{x^2+y^2}$ .Then, $\lim_{(x,y)\to(-12,5)}\sqrt{x^2+y^2}=13$

Example 19: Let $f(x,y)=\dfrac{x^2-y^2}{x^2+y^2}$.Then, the domain f is all points in the xy plane except the origin. We will approach the origin on a line, whose equation is $y=mx$. Then, we see that

$f(x,y)=f(x,mx)=\dfrac{x^2-(mx)^2}{x^2+(mx)^2}=\dfrac{x^2(1-m^2)}{x^2(1+m^2)}=\dfrac{1-m^2}{1+m^2}$. If the values m_1 and m_2 of the variable m satisfy the condition $m_1\neq m_2$, then the limits of $f(x,mx)$ at 0 are different. Finally, we conclude that f has no limit at the origin.

Example 20: Evaluate a) $\lim_{(x,y)\to(0,0)} \dfrac{x^3+y^3}{x^2+y^2}$ b) $\left(\lim_{(x,y)\to(0,0)} \dfrac{1-\cos(xy)}{x^2y^2+x^2y^3}\right)$

6: CONTINUITY OF FUNCTIONS OF TWO VARIABLES

Definition: **Let** $f : D \to \mathrm{R}$**, where** $D \subseteq \mathrm{R}^2$ **be a function and** $P_0 = (x_0, y_0) \in D$**. Then, f is said to be continuous at** $P_0 = (x_0, y_0)$

1. f Is defined at (x_0, y_0)
2. $\lim_{(x,y)\to(x_o,y_o)} f(x,y)$ exists
3. $\lim_{(x,y)\to(x_o,y_o)} f(x,y) = f(x_o,y_o)$

Theorem 3: Let f and g be continuous at P_0. Then

(i) $f \pm g$ is continuous at P_0.
(ii) fg is continuous at P_0.
(iii) cf is continuous at P_0, $c \in R$
(iv) $\dfrac{f}{g}$ is continuous at P_0. Provided that $g(p_0) \neq 0$

Theorem 4: Let f be a continuous function of two variables at the point P_0 .Then,

(i) If g is continuous real valued function at $f(P_0)$, then $g \circ f$ is continuous.

(ii) If h is a vector valued function in $\mathbf{R}^2$ which is continuous at t_0 and $h(t_0) = P_0$, then $f \circ h$ is continuous at t_0.

Theorem 5: Let f be a function of two variables such that f is defined in an open ball centered at $P_0 = (x_0, y_0)$.Then, f is continuous at P_0 if and only if

$$\lim_{P\to P_0} f(P) = f(P_0)$$

Example 21: The function defined by $f(x,y) = \dfrac{x^2+y^3}{x^3y+4}$ is continuous at the point $(0,0)$ because $\lim_{(x,y)\to(0,0)} f(x,y) = f(0,0)$.

Example22: Show that $f(x,y) = \begin{cases} (x^2+y^2)\ln(x^2+y^2) & if\ (x,y) \neq (0,0) \\ 0 & if\ (x,y) = (0,0) \end{cases}$ is continuous at $(0,0)$.

Solution Solution: on the board

Example23: Show that f is not continuous at $(0,0)$ where

$$f(x,y) = \begin{cases} \dfrac{\sin(xy)}{x^2+y^2} & if\ (x,y) \neq (0,0) \\ 0 & if\ (x,y) = (0,0) \end{cases}.$$

Solution: **Solution**: on the board

7: PARTIAL DERIVATIVES

Definition **Let** $f : \mathbf{D} \subseteq \mathrm{R}^2 \to \mathrm{R}$ **be a function of** x **and** y **and** $(x_0, y_0) \in \mathbf{D}$**. Then**

i. The partial derivative of f at (x_0, y_0) with respect to x is defined as
The partial derivative is denoted by

$$f_x(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}$$ (if it exists).

ii. The partial derivative of f at (x_0, y_0) with respect to y is given by

$$f_y(x_0, y_0) = \lim_{y \to y_0} \frac{f(x_0, y) - f(x_0, y_0)}{y - y_0} = \lim_{h \to 0} \frac{f(x_0, y_0 + h) - f(x_0, y_0)}{h}$$

Example 24: Let $f(x, y) = x^2 + y^2$. Find the partial derivatives of f at $(0,0)$.

Solution:

(i) By definition

$$f_x(0, 0) = \lim_{x \to 0} \frac{f(x, 0) - f(0, 0)}{x - 0} = \lim_{x \to 0} x = 0.$$

Hence, $f_x(0,0) = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=0 \\ y=0}} = 0$. Similarly, $f_y(0,0) = 0$.

Example 25: Consider $f(x, y) = \frac{x + y}{x - y}$. Then, we see that

$$\frac{\partial f(x, y)}{\partial x} = \frac{\partial}{\partial x}\left(\frac{x + y}{x - y}\right)$$

$$= \frac{1(x - y) - (x + y)(1)}{(x - y)^2} = \frac{-2y}{(x - y)^2}.$$

$$\frac{\partial f(x, y)}{\partial x} = \frac{\partial}{\partial x}\left(\frac{x + y}{x - y}\right)$$

$$= \frac{1(x - y) - (x + y)(-1)}{(x - y)^2} = \frac{2y}{(x - y)^2}.$$

Example 26: f(x, y) = $e^{xy^2} \cos y$. Then,

$$\frac{\partial f(x, y)}{\partial x} = \cos y(e^{xy^2})(y^2) = y^2(\cos y)(e^{xy^2}).$$

$$\frac{\partial f(x, y)}{\partial y} = (-\sin y)e^{xy^2} + 2xy \cos y e^{xy^2}.$$

$$= e^{xy^2}\left[-\sin y + 2xy \cos y\right].$$

Example 27: Find the partial derivatives of the following

a) $f(x,y)=e^{xy}+\dfrac{x}{y}$

b) $f(x,y)=\sqrt{9-x^2-y^2}$

c) $f(x,y,z)=(xy)^{\sin z}$

d) $f(x,y)=e^{x}\ln\left(x^2+y^2+1\right)$

e) $f(x,y,z)=z\sin^{-1}\left(\dfrac{y}{x}\right)$

8: HIGHER ORDER DERIVATIVES

Recall that for a function of two variables x and y, $\frac{\partial f}{\partial x}$and $\frac{\partial f}{\partial y}$ are functions of x and y.

By finding the partial derivatives of , $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$with respect to x and y, we obtain second order partial derivatives as defined below.

(i) The second order partial derivative with respect to the variable x is defined by

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)\Bigg|_{\substack{x=a\\y=b}}=\frac{\partial^2 f}{\partial x^2}\Bigg|_{\substack{x=a\\y=b}}$$ to denote $f_{xx}(a,b)$.

(ii) The second order partial derivative with respect to the variable y is defined by

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)\Bigg|_{\substack{x=a\\y=b}}=\frac{\partial^2 f}{\partial y^2}\Bigg|_{\substack{x=a\\y=b}}$$ to denote $f_{yy}(a,b)$.

(iii) The second order mixed partial derivatives at a point (a,b) are defined by

$$f_{xy}(a,b)=\lim_{y\to b}\frac{f_x(a,y)-f_x(a,b)}{y-b}$$ and $$f_{yx}(a,b)=\lim_{x\to a}\frac{f_y(x,b)-f_y(a,b)}{x-a}$$ provided that these limits exist. Here, f_{xy} and f_{yx} are denoted by $\frac{\partial^2 f}{\partial y\partial x}$and $\frac{\partial^2 f}{\partial x\partial y}$ respectively.

Example 28:

1. Find $f_{yx}, f_{xy}, f_{xx}, f_{yy}$

   a) $f(x,y)=2x^2y^3-3y^2x$

   b) $f(x,y)=\sqrt{x^2+y^2-1}$

   c) $f(x,y)=Cosh^2(xy)$

   d) $f(x,y)=\sin(x^2+y^3)$

   e) $f(x,y)=3\sin(x+y)-2\cos(x^2)$

   f) $f(x,y)=e^{x^2y}\cos(xy)$

2. Which of the mixed partial derivatives are equal in question number 1 ? Why?

3. Let $f(x,y)=e^{x^2y}$. Then, find f_{xy}, f_{yx}, f_{xyx} and f_{yxx}.

In general, the mixed partial derivatives may not be equal.

Theorem 5: Let f be a function of two variables x and y. If f_x, f_y, f_{xy}, f_{yx} are continuous in an open disk D, then $f_{xy}=f_{yx}$ at each point of the set D.

Example 29: find f_{xyy} *for* $f(x,y)=\cos(xy)-x^3+y^4$

9: CHAIN RULE

Consider a function f of one variable x and x is a function of variable t .Then, the chain rule states that $(f \circ x)'(t) = f'(x(t)).x'(t)$. In this section, we give the chain rule for function of two variables x and y .

- If we have a function $w = f(z)$ where $z = \varphi(x, y)$ and f is a differentiable function,

$$\Rightarrow \frac{\partial w}{\partial x} = \frac{dw}{dz} \cdot \frac{\partial z}{\partial x} \text{ and } \frac{\partial w}{\partial y} = \frac{dw}{dz} \cdot \frac{\partial z}{\partial y}.$$

Example 29: Suppose f is differentiable function of one variable $g(x, y) = f\left(\sqrt[3]{x^3 + y^2}\right)$

Find $\frac{\partial g}{\partial x}$ and $\frac{\partial g}{\partial y}$.

Solution: (Exercise)

Theorem 6 (First version of the chain rule for two variables)

Let $z = f(x, y)$ function of two variables differentiable at x and y Where $x = g(t)$ *and* $y = h(t)$ is differentiable ,then

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

Example 30: Let $z = x^2 + y^3$. Then, $\frac{\partial f}{\partial x} = 2x$, $\frac{\partial f}{\partial y} = 3y^2$. Suppose now that

$x = \cos(2t)$, $y = Cos(t)$. Then $\frac{dx}{dt} = -2\sin(2t)$, $\frac{dy}{dt} = -\sin t$. It follows that

$$\frac{df}{dt} = \frac{\partial f}{\partial x}.\frac{dx}{dt} + \frac{\partial f}{\partial y}.\frac{dy}{dt} = 2x(-2\sin(2t) + 3y^2(-\cos t) = 2\cos(2t)(-2\sin(2t) + 3\cos^2 t(-\sin t)$$

Example 31: Consider $f(x, y) = \log(x^2 + y^2)$, where $x(\theta) = \cos\theta$, $y(\theta) = \sin\theta$.

Then $\frac{df}{d\theta} = \frac{\partial f}{\partial x} \cdot \frac{dx}{d\theta} + \frac{\partial f}{\partial y} \cdot \frac{dy}{d\theta}$. But, we can see that

$$\frac{\partial f}{\partial x} = \left(\frac{1}{x^2 + y^2} \log e\right) \quad , \quad \frac{\partial f}{\partial y} = -\left(\frac{1}{x^2 + y^2}\right)(2y \log e) \quad .$$

$$\frac{dx}{d\theta} = -\sin\theta, \frac{dy}{d\theta} = \cos\theta$$

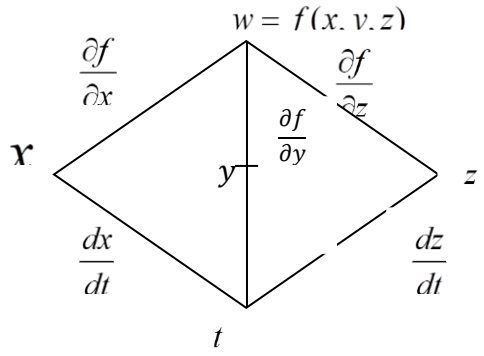
Example 32: Consider $f(x, y) = x^2 \cos y$, where $x(t) = t^2 + 2t$, $y(t) = \sin t$

Then, $\frac{df}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt} = (2x\cos y)(2t + 2) - (x^2 \sin y)(\cos t).$

Theorem 7: (First version of the chain rule for three variables):

Let *f* be a function of three variables *x*, *y*, *z*, differentiable on an open disk centered disk D. Suppose that *x*, *y* and *z* are functions of a single variable *t*, differentiable on an open interval I, and such that for every $t \in I, (x(t), y(t), z(t)) \in D$. Then $w = f(x(t), y(t), z(t))$

is a function of *t*, differentiable on I and $\frac{df}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial f}{\partial z} \cdot \frac{dz}{dt}$



Example 33: Let $w = \cos(x^2 z^3 + y) + xy,\ x(t) = 2t,\ y(t) = t^2 \sin t, z(t) = te^{t^2}$. Find $\frac{dw}{dt}$

Solution: on the board

Theorem 8 (The second version of the chain rule): If $x = x(u,v)$ and $y = y(u,v)$ have first partial derivatives at (u,v) and $z = f(x,y)$ is differentiable at $(x(u,v), y(u,v))$, then

(i) $$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}.\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}.\frac{\partial y}{\partial u}$$

(ii) $$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}.\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}.\frac{\partial y}{\partial v}$$

Example 34: Consider $f(x,y) = x^2 - y\tan x$ where $x = u^2 + v^2$, $y = u + v$

Find $\frac{\partial f}{\partial u}, \frac{\partial f}{\partial v}$

Solution: on the board



Figure 4.3
The second version of the chain rule

10: IMPLICIT FUNCTIONS OF THREE VARIABLES

- Suppose y and x are variables such that y is a function of x such that y can't be explicitly expressed as a function of x. Suppose f is differentiable function of x and y such that $f(x,y) = 0$. Then, we need to find $\frac{dy}{dx}$ at all points at which y is differentiable.

By applying the chain rule, we get $\frac{\partial f}{\partial x}.\frac{dx}{dx}+\frac{\partial f}{\partial y}.\frac{dy}{dx}=0$.

$$\frac{dy}{dx}=-\frac{f_x}{f_y}$$

Example 35: Find y' *if* $x^3+y^3=6xy$

Solution: $f(x,y)=x^3+y^3-6xy=0$

$$\frac{dy}{dx}=-\frac{f_x}{f_y}=-\frac{3x^2-6y}{3y^2-6x}=\frac{2y-x^2}{y^2-2x}$$

- Assume that the variables *x*, *y* and *z* are related by the equation *F*(*x*, *y*, *z*) = 0. Assuming that $F_z \neq 0$, there is a version of the implicit function theorem which means, we can in principle, write *z* = *z*(*x*, *y*), so we can solve for z. Thus, consider the equation $\omega =$ *F*(*x*, *y*, *z*(*x*, *y*)) = 0 and differentiating both sides partially with respect to *x*. We get

$\frac{\partial \omega}{\partial x}=\frac{\partial F}{\partial x}.\frac{\partial x}{\partial x}+\frac{\partial F}{\partial y}.\frac{\partial y}{\partial x}+\frac{\partial F}{\partial z}.\frac{\partial z}{\partial x}=0.$ and since $\frac{\partial F}{\partial z}\neq 0$, we get

$$\Rightarrow \frac{\partial z}{\partial x}=-\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}}=\frac{-F_x}{F_z}.$$

$$\frac{\partial \omega}{\partial y}=\frac{\partial F}{\partial x}.\frac{\partial x}{\partial y}+\frac{\partial F}{\partial y}.\frac{\partial y}{\partial y}+\frac{\partial F}{\partial z}.\frac{\partial z}{\partial y}=0.$$

$$\Rightarrow \frac{\partial z}{\partial y}=-\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}}=\frac{-F_y}{F_z}.$$

Example 36: Find $\frac{\partial z}{\partial x},\frac{\partial z}{\partial y}$ if (i) $x^2+y^2+z^2=9$ (ii) $\sqrt{xy+z^3}=\cos(yz^2)$

Solution: (ii) Let $F(x,y,z)=\sqrt{xy+z^3}-\cos(yz^2)$. Then,

$$F(x,y,z)=0 \Leftrightarrow \sqrt{xy+z^3}=\cos(yz^2).$$

Differentiating both sides of the equation $\sqrt{xy+z^3}=\cos(yz^2)$ with respect to y, we get

$$\frac{1}{2\sqrt{xy+z^3}}(x+3z^2\frac{\partial z}{\partial y})=-\sin(yz^2).\left[z^2+2yz\frac{\partial z}{\partial y}\right] \text{ which is equivalent to}$$

$$\left(\frac{3z^2}{2\sqrt{xy+z^3}}+2yz\sin(yz^2)\right)\frac{\partial z}{\partial y}=-z^2\sin(yz^2)-\frac{x}{2\sqrt{xy+z^3}}.$$

Thus, solving for $\frac{\partial z}{\partial y}$, we get $\frac{\partial z}{\partial y}=\frac{-\left(x+2z^2\sin(yz^2)\sqrt{xy+z^3}\right)}{\left(4yz\sin(yz^2)\right)\sqrt{xy+z^3}+3z^2}$.

One can check the result using the formula $\frac{\partial z}{\partial y} = -\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}} = \frac{-F_y}{F_z}$.

11: THE GRADIENT AND DIRECTIONAL DERIVATIVES

Definition 5.1.1: Let f be a function of two variables that has partial derivatives at (x_0, y_0). Then, the gradient of f at (x_0, y_0) is defined as $\nabla f(x_0, y_0) = f_x(x_0, y_0)i + f_y(x_0, y_0)j$.

Now, we give the definition of directional derivatives for a function of two variables as follows.

Definition 5.1.2: Let f be a function defined on a set containing a disk D centered at (x_0, y_0) and $u = a_1 i + a_2 j$ be a unit vector. The directional derivative of f at (x_0, y_0) in the direction of u, denoted by $D_u f(x_0, y_0)$ is defined as

$$D_u f(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0 + ha_1, y_0 + ha_2) - f(x_0, y_0)}{h}$$

provided that this limit exists.

- ✓ If $u = (1, 0)$, then

$$D_u f(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h} = f_x(x_0, y_0)$$

- ✓ If $u = j = (0,1)$, then

$$D_u f(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0, y_0 + h) - f(x_0, y_0)}{h} = f_y(x_0, y_0).$$

Thus, $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ are special kind of directional derivatives in the direction of $i = (1,0)$ and $j = (0,1)$ respectively.

Example 37

Find the directional derivative of the following functions at the indicated points.

(a) $f(x, y) = 2x^2 - 3xy + y^2 + 15$ at(1, 1) in the direction of $u = \frac{1}{\sqrt{2}} i + \frac{1}{\sqrt{2}} j$.

(b) $f(x, y) = x^2 + y^2$ at (1, 2) in the direction of $u = \frac{1}{\sqrt{3}} i - \frac{\sqrt{2}}{\sqrt{3}} j$.

(c) $f(x, y, z) = 3x - 2y + 4z$ at $(1,-1,2)$ in the direction of $u = i + j$.

(d) $f(x, y) = y^2 \ln x$ at(1, 4) in the direction of $u = -3i + 3j$.

(e) $f(x, y) = e^x \cos y$ at $(0, \frac{\pi}{4})$ in the direction of $u = 5i - 2j$

Now, the following theorem will show that the directional derivative of a function exists in any direction provided that the limit exists.

Theorem 10: Suppose f is differentiable at (x_0, y_0). Then f has directional derivative at any direction. Hence, $D_u f(x_0, y_0) = \nabla f(x_0, y_0) \bullet u$

Remark: Let f be a function of three variables x, y and z. Then,

(a) $\nabla f(x_0, y_0, z_0) = f_x(x_0, y_0, z_0)i + f_y(x_0, y_0, z_0)j + f_z(x_0, y_0, z_0)k$

(b) For any vector $u = a_1 j + a_2 j + a_3 k$ (u is unit vector),

$$D_u f(x_0, y_0, z_0) = \nabla f(x_0, y_0.z_0) \bullet u.$$

Example 38: let $f(x, y, z) = x\sin(yz)$, *find the* directional derivative of f at (1,3,0) in the direction of v=i+2j-k.

Solution : $D_v f(1,3,0) = \frac{v}{\|v\|} \bullet \nabla f(1,3,0)$

Theorem 11.: Suppose f is differentiable at (x_0, y_0). Then

(a) $\nabla f(x_0, y_0) = 0$ implies $D_u f(x_0, y_0) = 0$ for any vector u .(No rate of change)

(b) If $\nabla f(x_0, y_0) \neq 0$, then

- ✓the directional derivative of f in the direction of $\nabla f(x_0, y_0)$ is the largest and its value is $\|\nabla f(x_0, y_0)\|$. The maximum rate of change of f at (x_o, y_o) occurred when u and $\nabla f(x_0, y_0)$ have the same direction.
- ✓ The minimum rate of change is -$\|\nabla f(x_0, y_0)\|$ which occur when u and $\nabla f(x_0, y_0)$ have the opposite direction.

Example 39: in which direction does the function $f(x, y) = xy^2 + x^3 y$ increase the fastest at the point (1,2)? In which direction it decrease fastest.

Solution:

(Exercise)

12:Total Differential

For a differentials of two variables $z = f(x, y)$, we define the differentials dx and dy and to be independent variables; i.e. they can be given any values. Then the differential, also called the total differential, is defined by

$$dz = f_x(x, y)dx + f_y(x, y)dy$$
$$= \frac{\partial z}{\partial x}(x, y)dx + \frac{\partial z}{\partial y}(x, y)dy$$

Let $f(x, y) = x^4 y^6$. Find the total differential

Solution:

Since $f(x, y) = x^4 y^6$

$f_x(x, y) = 4x^3 y^6$ and $f_y(x, y) = 6x^4 y^5$

So $dz = f_x(x, y)dx + f_y(x, y)dy = 4x^3 y^6 dx + 6x^4 y^5 dy$

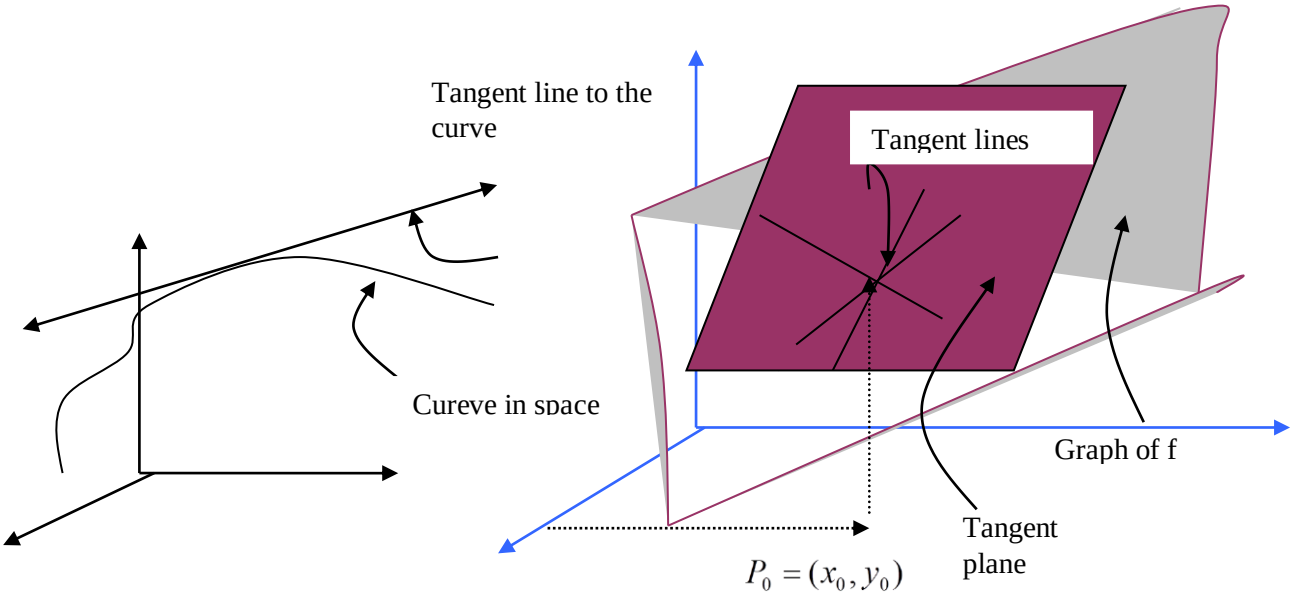
Exercise: Find the total differential of

a) $g(x, y) = y\cos xy$
b) $h(x, y, z) = xz\sin y + z^2$

13: TANGENT PLANES

In this subtopic, we will find the equation of a tangent plane to the graph of a function $z = f(x, y)$ at a given point $(x_0, y_0, f(x_0, y_0))$ and to the surface $f(x, y, z) = c$ where f is a function of x and y. For a function f of one variable, if f is differentiable at c, we know that $y = f(c) + (x - c)f'(c)$ is defined to be the equation of the tangent line to the graph of f at the

point $(c, f(c))$. Analogously, we define the equation of the tangent plane to the graph of f of two variables as follows. See the following graphical illustrations.



Definition: Suppose f is differentiable at $P_0 = (x_0, y_0)$.Then, the equation of the tangent plane to the graph of f at $(P_0, f(P_0))$ is given by $z = f(P_0) + (P - P_0) \bullet \nabla f(P_0)$

or $f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - z = 0$

Example 40: find the equation of the tangent plane to the elliptical paraboloid $z = 3x^2 + 2y^2$ *at* $(1,1,5)$

Solution

Theorem 12: The equation of the tangent plane to the surface $f(x, y, z) = c$ at the point (x_0, y_0, z_0) is given by

$$\frac{\partial f(x,y,z)}{\partial x}(x - x_0) + \frac{\partial f(x,y,z)}{\partial y}(y - y_0) + \frac{\partial f(x,y,z)}{\partial z}(z - z_0) = 0.$$

Example 41: Find the equation of the tangent plane to the surface $f(x, y, z) = x^2 + y^2 + z - 9 = c$ at the point $P = (1,2,4)$.

Solution: We have $\nabla f(1,2,4) = (2,4,1)$, and the equation of the tangent plane is

$$2(x-1) + 4(y-2) + (z-4) = 0 .$$

[14: Tangent Plane Approximation of Values of a Function Reading Assignment]

15: EXTREMA OF FUNCTIONS OF TWO VARIABLES

Definition: Let f be a function of two variables. Then,

(a) f is said to have relative maximum value at (x_0, y_0) if and only if there exists a disk centered at (x_0, y_0) such that $f(x, y) \le f(x_0, y_0)$ for all (x, y) in the interior of the disk.

(b) f is said to have relative minimum value at (x_0, y_0) if and only if there exists a disk centered at (x_0, y_0) and $f(x,y) \geq f(x_0, y_0)$ for all (x,y) in the disk.

(c) f is said to have absolute maximum at (x_0, y_0) if and only if $f(x_0, y_0) \geq f(x,y)$ for all (x,y) in the domain of f.

(d) f is said to have absolute minimum at (x_0, y_0) if and only if $f(x_0, y_0) \leq f(x,y)$ for all (x,y) in the domain of f.

Theorem 13: If $f(x,y)$ is continuous in a *closed and bounded* subset S of the plane, then f has both absolute maximum/minimum/value in S.

Theorem 14: If f has relative extremum at (x_0, y_0) and $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ exist, then $f_x(x_0, y_0) = f_y(x_0, y_0) = 0.$

Definition: Suppose f is a function of two variables. Let (x_0, y_0) be in the domain of f. Then (x_0, y_0) is called the **critical point** of f if and only if $f_x(x_0, y_0) = f_y(x_0, y_0) = 0$ or either $f_x(x_0, y_0)$ *or* $f_y(x_0, y_0)$ doesn't exist.

Example 42 Let $f(x,y) = xy.$ Then $f_x = y$ and $f_y = x$. We have

$f_x = f_y = 0 \Leftrightarrow x = y = 0.$ The function *f* has a single critical point, namely the origin.

In any disk centered at the origin, the function attain minimum through the line $y = x.$ Moreover, the function attains maximum at $(0,0)$ through the line $y = -x.$

Hence, $(0,0)$ is a saddle point of f. (See the figure below).

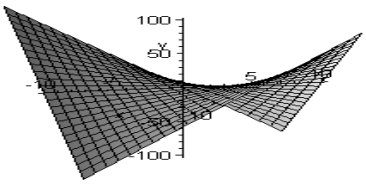


Figure 5.3
Graph of $z = xy$

Example 43: Let $f(x,.y) = x^2 y + x - y$. Then, we get

$$f_x(x,y) = 2xy + 1 = 0 \text{ and } f_y(x,y) = x^2 - 1 = 0$$

Solving these equations, we obtain $x = \pm 1, y = \pm\frac{1}{2}$ .The function *f* has four critical points, namely $(1, \frac{-1}{2}), \left(-1, \frac{1}{2}\right), \left(-1, \frac{-1}{2}\right)$ and $\left(1, \frac{1}{2}\right)$.

Theorem 15: (Second partial derivative test): Let f be a function of two variables with continuous second-order partial derivatives in some disk centered at the critical point (x_0, y_0).

Let $D = \begin{vmatrix} f_{xx}(x_0, y_0) & f_{xy}(x_0, y_0) \\ f_{yx}(x_0, y_0) & f_{yy}(x_0, y_0) \end{vmatrix} = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0).$

(a) If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then f has relative minimum at (x_0, y_0)

(b) If $D > 0$ and $f_{xx}(x_0, y_0) < 0$, then f has relative maximum at (x_0, y_0)

(c) If $D < 0$, then f has saddle point at (x_0, y_0)

(d) If $D = 0$, no conclusion can be drawn.

Example 44: Consider $f(x, y) = x^2 + xy - 2y - 2x + 1$. Find all the critical points and determine the relative maximum and relative minimum values of f.

Solution: **Solution**: on the board

Example 45: Locate all the critical points and find all the points of relative maximum and relative minimum of $f(x, y) = x^3 + xy + y^3$.

Solution: Let (x, y) be a relative extremum point of f. Then, we see that

$$f_x(x, y) = 3x^2 + y = 0, \quad f_y(x, y) = 3y^2 + x = 0$$

Thus, solving for y in terms of x and substitution yields $27x^4 + x = 0$ .This is true if

$x = 0$ or $x = \frac{-1}{3}$, i.e. $(0,0)$ and $\left(\frac{-1}{3}, \frac{-1}{3}\right)$ are the only critical points of f.

Now, we obtain

$f_{xx}(0,0) = 0,\ f_{xx}(\frac{-1}{3}, \frac{-1}{3}) = -2,\ f_{yy}(0,0) = 0,\ f_{yy}\left(\frac{-1}{3}, \frac{-1}{3}\right) = -2,\ f_{xy}(x, y) = 1 = f_{yx}(x, y)$

and $D(0,0) = -1 < 0,\ D\left(\frac{-1}{3}, \frac{-1}{3}\right) = 3 > 0.$ By the second partial test, we conclude that $(0,0)$ is a saddle point and $\left(\frac{-1}{3}, \frac{-1}{3}\right)$ is a point of relative maximum

Example 46: Let $f(x, y) = 2x^3 - 6x^2 - 3y^2 - 6xy$. Find and classify the critical points of *f*. By considering *f* (*x*, 0), or otherwise, show that *f* does not achieve a global maximum.

Solution: We have $f_x(x, y) = 6x^2 - 12x - 6y$ and $f_y(x, y) = -6y - 6x$. Thus critical points occur when $y = -x$ and $x^2 - x = 0$, and so at (0, 0) and (1, - 1). Differentiating again, $f_{xx}(x, y) = -6,\ f_{yy}(x, y) = -6$ .Thus the discriminator is $D(0,0) = 36 > 0$ and $D(1,-1) = -36 < 0$. We have a local maximum at (0, 0) and a saddle point at $(1,-1)$.

Example 47: Find a point on the plane $2x - 3y - 4z = 25$ closest to the point $(3,2,1)$.

Solution: on the board

16: EXTREME VALUES IN A BOUNDED REGION

Consider a function f that is continuous on a bounded subset S of the *xy* plane. Then, by extremum value theorem, f attain its absolute maximum and absolute minimum in S. Thus, the absolute maximum/minimum/ is obtained either on the boundary of S or in the interior of S. Now, we led to the following steps to find absolute extremum values of f in a bounded region S.

Step 1: Find critical points that lie in the interior of S.

Step 2: Obtain the maximum/minimum/ value of f on the boundary of S.

Step 3: Compare the values of f obtained in step 1 and step 2.

Example 48: Find the extreme values of $f(x,y) = x^2 - y^2$ on S where $S = \{(x,y) : x^2 + y^2 \le 1\}$.

Solution: $f_x(x,y) = 0 \Leftrightarrow x = 0$ and $f_y(x,y) = 0 \Leftrightarrow y = 0.$

Thus, (0, 0) is the only critical point of f in the circular region S. Now, let (x, y) be on the boundary of R. Then, $x^2 + y^2 = 1$, i. e. $y = \pm\sqrt{1-x^2}$.

Thus, $g(x) = 2x^2 - 1.$ By applying the second derivative test for $g(x) = 2x^2 - 1,$ we obtain that $g(0) = -1$ is minimum value of g. On the other hand, $f(0,\pm 1) = -1$. Thus, comparing the values $f(0,0) = 0$ and $f(0,\pm 1) = -1$, we conclude that $f(0,0) = 0$ is absolute maximum of f in $S.$ Moreover, $f(0,\pm 1) = -1$ is minimum value in S.

Example 49: A rectangular box with volume 16 cm^3 is made from two kinds of material. The top and the bottom are made of a material costing 10 cents/ sq.cm and the sides from a material costing 5 cents/sq. cm. Find the dimensions of a box with minimum cost.

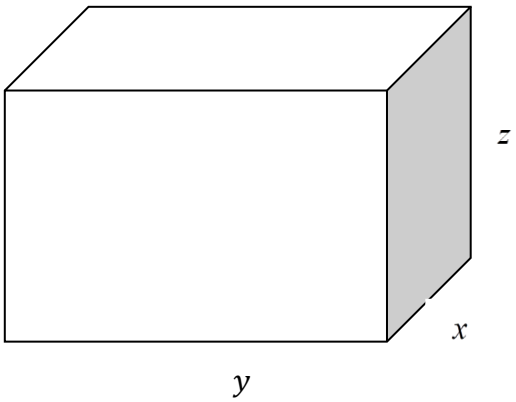


Figure 5.9

Let $v(x,y,z) = xyz = 16$ .The bottom and the top of the box costs $10(2xy) = 20(xy)$. Moreover, the sides cost $5(2xz) = 10(xz)$ and $5(2yz) = 10(yz)$.

Thus, the cost of the box is given by

$$C(x,y,z) = 20xy + 10xz + 10yz$$

Since $xyz = 16$, we see that z = $\frac{16}{xy}$. Thus, we get c(x, y) = 20xy + $\frac{160}{y} + \frac{160}{x}$.

Now, we find the critical points using the 1nd partial derivative .

i.e. $\frac{\partial c}{\partial x} = 20y - \frac{160}{x^2} = 0$ and $\frac{\partial c}{\partial y} = 20x - \frac{160}{y^2} = 0$. Solving for the variable x and substituting, we get $8y - y^4 = 0$ which gives $y = 0$ or $y = 2$.Thus, we obtain $x = 2.$ By using the second partial test, it is possible to check that $(2,2)$ is a point of relative minimum. Hence, the least dimension of the box with minimum cost is $x = 2, y = 2$ and $z = 4.$

Example 50: An open-topped rectangular tank is to be constructed so that the sum of the height and the perimeter of the base is 30 meters. Find the dimensions which maximize the surface area of the tank. What is the maximum value of the surface area?

Solution: on the board

Example 51: Consider $f(x,y) = xy - y + x - 1$ .Find absolute maximum/minimum/ of f in the closed disk $D = \{(x,y) : x^2 + y^2 \le 4\}$..

Solution:

$$f_x(x,y) = 0 \Leftrightarrow y = -1 \text{ and } f_y(x,y) = 0 \Leftrightarrow x = 1.$$

Then, (1, -1) is the only critical point of f in the interior of the disk D. One can show that (1, -1) is a **saddle point**. Then, we search the maximum/minimum/ on the boundary of D. i.e. on the circle $x^2 + y^2 = 4$. One of the parametric equation of $x^2 + y^2 = 4$ is $x(t) = 2\cos t, y(t) = 2\sin t, 0 \le t \le 2\pi$..

Thus, by substituting x and y in $(x,y) = xy - y + x - 1$, we get

$f(x,y) = f(t) = 4\sin t \cos t - 2\sin t + 2\cos t - 1$. Thus, by the second derivative test, one can show that f assumes maximum at

$$\left(\frac{1+\sqrt{3}}{\sqrt{2}}, \frac{\sqrt{3}-1}{\sqrt{2}}\right) and \left(\frac{1-\sqrt{3}}{\sqrt{2}}, -\frac{\sqrt{3}+1}{\sqrt{2}}\right).$$

Moreover, f assumes minimum at.$(-\sqrt{2}, \sqrt{2})$

17: LAGRANGE'S MULTIPLIER

One of the applications of the implicit function theorem is the method of finding extreme values of a function f on the contour line of another function. In this topic, we introduce the procedure called **Lagrange Multiplier Method**.

Theorem: Suppose f and g have continuous derivatives throughout a disk centered at $P_0 = (x_0, y_0)$ such that $g(P_0) = c$ and that $f(P_0) \ge f(P)$ for all points on the contour line $\{(x,y) : g(x,y) = c\}$. If $\nabla g(P_0) \ne 0,$ then there exists λ such that $\nabla f(P_0) = \lambda \nabla g(P_0)$.

The number λ is called the Lagrange's Multiplier.

Example 52: Find the maxima and minima of the function $f(x,y) = x^2 + y^2$ on the ellipse $x^2 + 2y^2 = 1$.

Solution: Let $g(x,y) = x^2 + 2y^2$ and suppose (x, y) is a point of extreme value of f on $x^2 + 2y^2 = 1$.Then, there exists a real number λ such that

$$\nabla f(x,y) = \lambda \nabla g(x,y).$$

This is equivalent to solving

$$\begin{cases} 2x = \lambda(2x) \\ 2y = \lambda(4y) \end{cases} \quad \text{for } x, y \text{ and } \lambda.$$

Since (x,y) is on the ellipse, both x and y can't be zero at the same time. Hence, we get either $x = 0, \lambda = \frac{1}{2}$ or $y = 0, \lambda = 1$. Since we seek points on the ellipse, $x = 0$ implies $y = \pm\frac{1}{\sqrt{2}}$ and

$y = 0$ implies that $x = \pm 1$. Moreover, we see that $f(0,\pm\frac{1}{\sqrt{2}}) = \frac{1}{2}$, $f(\pm 1,0) = 1$. We conclude that if f has maximum, then it occurs at $(\pm 1,0)$ and if it has minimum, then it occurs at $(0,\pm\frac{1}{\sqrt{2}})$.